

A semi-synthetic benchmark with a known exposure–response curve

Why the construction is what it is, and what it does and does not guarantee

1 The problem

We want to know whether the model recovers a causal exposure–response curve. On real data this question has no answer: the true curve is exactly the unknown being estimated, so a recovered curve can only be compared against another estimate, and any disagreement is uninterpretable.

The way out is to build data whose answer we chose. We keep the *real* covariates — real PM_{2.5}, real gridMET, real census — and synthesise only the *outcome*, from a rate we write down ourselves. The model trains on exactly the numbers that produced its target, and we can ask whether the curve it recovers is the curve we injected.

This is worth doing only if two things hold. The benchmark must be **hard**: an unconfounded exposure would let naive regression recover the truth, failing to separate a causal estimator from a curve-fitter. And the ground truth must be **exact**: a Monte-Carlo “truth” carries its own error bars, hiding small biases. Sections 2 and 3 address these in turn.

2 The generative model

Let t index days, i index the V ZCTAs. Write $X_{t,i}$ for the exposure (PM_{2.5}, in $\mu\text{g}/\text{m}^3$) and $\tilde{C}_{t,i}^{(k)}$ for the k -th standardised confounder. The per-capita Poisson rate is a sum of terms we specify:

$$\eta_{t,i} = r_0 + \underbrace{A_s \sin\left(\frac{2\pi t}{365.25}\right) + \ell_{\text{lat}}(i) + \ell_{\text{lon}}(i)}_{\text{seasonal + geographic background}} + \beta X_{t,i} + \sum_k \gamma_k \tilde{C}_{t,i}^{(k)}, \quad (1)$$

$$\lambda_{t,i} = \max(\lambda_{\min}, \eta_{t,i}), \quad Y_{t,i} \sim \text{Poisson}(\lambda_{t,i} o_i), \quad o_i = \kappa \cdot \text{pop}_i. \quad (2)$$

Here β is the ground-truth dose–response and each γ_k is a known confounder effect. The floor $\lambda_{\min}$ keeps the rate positive; o_i is the population offset, so λ is a rate per person and λo an expected count.

Three choices here are deliberate. The link is **identity, not log**, so $\mathbb{E}[Y]/o$ is exactly linear in X : the ground truth is a closed form rather than an integral, and an ordinary least-squares fit recovers β and every γ_k directly — a check on the generator that needs no model at all. The **exposure enters raw and confounders standardised**, so β is “per $\mu\text{g}/\text{m}^3$ ” and comparable to the epidemiological literature, while each γ_k is per standard deviation and so comparable across covariates on very different scales. And the dose–response is **same-day** — $\lambda_{t,i}$ depends on $X_{t,i}$ and nothing else — the simplest defensible choice, with a sharp consequence for windowed forecasting (Section 4).

Closing the confounding triangle

Confounding needs two arms: the confounder must affect the outcome, *and* it must affect the exposure. Equation (1) supplies the first through γ_k . The second is whatever the real data happens to contain — real income is correlated with real PM_{2.5}, but by an amount we neither control nor know.

An optional generator supplies the missing arm on demand, decomposing each exposure into three independent unit-variance parts,

$$X_e = m_e + s_e \left(\sqrt{f^{\text{dr}}} z\left(\sum_k w_{ek} \tilde{C}^{(k)}\right) + \sqrt{f^{\text{st}}} z(\mathcal{S}_{\rho,\phi}\varepsilon) + \sqrt{1 - f^{\text{dr}} - f^{\text{st}}} z(\varepsilon') \right), \quad (3)$$

where $z(\cdot)$ standardises and $\mathcal{S}_{\rho,\phi}$ imposes spatial and temporal correlation. Because the parts are orthogonal by construction, R^2 of the exposure on the confounders equals f^{dr} — the confounding strength becomes a dial, not an accident.

The residual fraction $f^{\text{no}} = 1 - f^{\text{dr}} - f^{\text{st}}$ is not slack: it is the exposure variation remaining *conditional on* the confounders. Driving it to zero destroys positivity and makes the curve unidentifiable at any sample size, so the generator refuses it.

Arm	Mechanism	Known?
confounder $\rightarrow$ outcome	γ_k in (1)	set in config
exposure $\rightarrow$ outcome	β in (1)	set in config; the target
confounder $\rightarrow$ exposure	f^{dr} in (3)	set in config <i>when generated</i>

Both worlds are supported. With *real* exposures the third arm is uncontrolled but the data are more realistic and the model needs no reconfiguration; with *generated* exposures the third arm is exact. The first is the default; one config switch selects the other.

3 The ground truth

Two interventions are of interest. The **level** intervention $\text{do}(X = a)$ sets the exposure to a constant everywhere and traces the classical exposure–response curve. The **shift** intervention $X \mapsto \delta X$ rescales the observed field, preserving its spatial and temporal texture; $\delta = 1$ is the factual world. Under either, confounders are left at their observed values — that is precisely the g-computation operator.

The crucial point is that (1) is deterministic given the covariates. The Poisson draw in (2) adds noise to the realised counts but not to their *mean*, and the ground truth is a statement about means. So we never simulate: we evaluate λ under the intervention directly, in closed form, including the floor. For the unfloored cells the slopes are analytic,

$$\frac{\partial}{\partial a} \mathbb{E}[Y \mid \text{do}(X = a)] = \beta \circ, \quad \frac{\partial}{\partial \delta} \mathbb{E}[Y \mid X \mapsto \delta X] = \beta \circ \mathbb{E}[X], \quad (4)$$

so the two curves differ only by the exposure’s own magnitude — a cross-check we use to validate them against each other. The floor is the only source of inexactness: it is a genuine nonlinearity, so (4) holds exactly only where $\eta > \lambda_{\min}$, which in practice is about 99% of cells.

Units are part of the estimand. The data loader serves every covariate standardised. Multiplying that channel by δ therefore implements $X \mapsto \mu + \delta(X - \mu)$, a fan-out about the mean, not $X \mapsto \delta X$. The two are different interventions and can even have opposite sign: if the year’s mean exposure sits below the normalising mean μ , “increasing by 10%” in standardised units *decreases* the expected outcome. Whichever convention the evaluation uses, the ground truth must use the same one.

4 The estimand is a sum over forecast windows

The model is not a per-day regressor. From a start day t_0 it consumes W daily input slices ($\tau = 0, \dots, W-1$, calendar day $t_0 + \tau$) and emits H forecast leads off each ($k = 0, \dots, H-1$). All leads come from one forward pass — a learned per-lead embedding on a shared latent — so the model is *direct multi-horizon*, not autoregressive. Nothing is fed back, so no error compounds and the closed form of Section 3 survives intact.

Prediction (t_0, τ, k) targets calendar day $s = t_0 + \tau + k$. Sweeping t_0 across the year and summing gives the estimand

$$G(\delta) = \frac{1}{V} \sum_i \sum_{t_0} \sum_{\tau=0}^{W-1} \sum_{k=0}^{H-1} f(\mathbf{X}^{(t_0)}, A^{(t_0)}[m^* \mapsto \delta A_{m^*}^{(t_0)}])_{\tau, k, i}. \quad (5)$$

Two distinct multiplicities follow, and conflating them is the easiest way to misread a result.

First, a day in the interior of the year is summed $W \cdot H$ times, not H times. A correction of “ $\times H$ because the model predicts a window” is short by a factor of W .

Second — and this is the subtle one — the intervention $A^{(t_0)}$ touches only that sample’s own W input days, $t_0, \dots, t_0 + W - 1$. A prediction with $\tau + k \geq W$ therefore targets a day whose exposure was left factual. Because the dose–response is same-day, those predictions carry *exactly zero* causal contrast, however large δ is. Writing $w_{\text{all}}(s)$ and $w_{\text{int}}(s)$ for the two counts,

$$G(\delta) = \frac{1}{V} \sum_i o_i \sum_s \left[w_{\text{int}}(s) \lambda_{s,i}(\delta) + (w_{\text{all}}(s) - w_{\text{int}}(s)) \lambda_{s,i}(1) \right], \quad (6)$$

with $w_{\text{int}}/w_{\text{all}} = \frac{1}{2}(W+1)/H$ in the interior (both ramp at the year’s edges). At $W = H = 30$ that is $465/900 \approx 52\%$; at $W = 3, H = 5$ it is $6/15 = 40\%$. The remainder dilutes the curve without contributing signal.

The practical consequence: a perfectly calibrated model scored against a ground truth that intervened on the *whole* exposure field, rather than window-locally, appears to recover only that fraction of the true effect. Both conventions are legitimate — they are different estimands — but they must match on both sides, and W and H must be the values the model was trained with.

5 What this does and does not give you

The construction guarantees that β and every γ_k are known exactly; that the ground truth is computed from the same rate expression the generator draws from, so the two cannot drift; and that a manifest of the resolved parameters is written beside every dataset, so the answer key travels with the data rather than with a config file that may have moved on.

It does not make the benchmark realistic in every respect. The outcome is additive and same-day by construction, so it rewards estimators matched to that structure. Poisson noise comes from a single seeded stream, so residuals are correlated across diseases and years and should not be read as signal. A confounder that is constant in the source data contributes nothing regardless of its stated γ — harmless for identification, but not a variable the model must adjust for. And the benchmark certifies that a correct adjustment recovers β ; it says nothing about robustness to confounders outside the specified set.